```python
import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
True
```

```python
import pandas as pd

data = {
    "sentence": ["I am happy","She is beautiful",
                  "He likes cricket",
                  "They are playing football",
                   "We have completed the work","I am not happy",
                  "She is not beautiful",
                  "He does not like cricket",
                  "They are not playing football",
                  "We have not completed the work"
    ],
    "sentiment": [
        "positive", "positive", "positive", "positive", "positive",
        "negative", "negative", "negative", "negative", "negative"
    ]
}

df = pd.DataFrame(data)
df
```

	sentence	sentiment
0	I am happy	positive
1	She is beautiful	positive
2	He likes cricket	positive
3	They are playing football	positive
4	We have completed the work	positive
5	I am not happy	negative
6	She is not beautiful	negative
7	He does not like cricket	negative
8	They are not playing football	negative
9	We have not completed the work	negative

Next steps: Generate code with df New interactive sheet

```
import nltk
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
nltk.download('punkt_tab') # Download the missing resource
stop_words = set(stopwords.words("english"))
ps = PorterStemmer()
lemm = WordNetLemmatizer()
def preprocess(text):
    text = text.lower()
    tokens = nltk.word_tokenize(text)
    no_stop = [w for w in tokens if w not in stop_words]
    stemmed = [ps.stem(w) for w in no_stop]
    lemmatized = [lemm.lemmatize(w) for w in no_stop]
    return tokens, no_stop, stemmed, lemmatized

df["tokens"] = df["sentence"].apply(lambda x: preprocess(x)[0])
df["no_stopwords"] = df["sentence"].apply(lambda x: preprocess(x)[1])
df["stemmed"] = df["sentence"].apply(lambda x: preprocess(x)[2])
df["lemmatized"] = df["sentence"].apply(lambda x: preprocess(x)[3])

df
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
```

	sentence	sentiment	tokens	no_stopwords	stemmed	lemmatized
0	I am happy	positive	[i, am, happy]	[happy]	[happi]	[happy]
1	She is beautiful	positive	[she, is, beautiful]	[beautiful]	[beauti]	[beautiful]
2	He likes cricket	positive	[he, likes, cricket]	[likes, cricket]	[like, cricket]	[like, cricket]
3	They are playing football	positive	[they, are, playing, football]	[playing, football]	[play, footbal]	[playing, football]
4	We have completed the work	positive	[we, have, completed, the, work]	[completed, work]	[complet, work]	[completed, work]
5	I am not happy	negative	[i, am, not, happy]	[happy]	[happi]	[happy]
6	She is not beautiful	negative	[she, is, not, beautiful]	[beautiful]	[beauti]	[beautiful]
7	He does not like cricket	negative	[he, does, not, like, cricket]	[like, cricket]	[like, cricket]	[like, cricket]
8	They are not playing football	negative	[they, are, not, playing, football]	[playing, football]	[play, footbal]	[playing, football]
9	We have not completed the work	negative	[we, have, not, completed, the, work]	[completed, work]	[complet, work]	[completed, work]

Next steps: Generate code with df New interactive sheet

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB

cv = CountVectorizer()
X = cv.fit_transform(df["sentence"])
y = df["sentiment"]

model = MultinomialNB()
model.fit(X, y)

print("Model training completed!")
```

```
Model training completed!
```

```
test_data = ["I am happy","She is beautiful",
                  "He likes cricket","He does not like cricket",
                  "They are not playing football",
                  "We have not completed the work"
```

```
]

X_test = cv.transform(test_data)
pred = model.predict(X_test)

for t, p in zip(test_data, pred):
    print(f"{t} --> {p}")
```

```
I am happy --> positive
She is beautiful --> positive
He likes cricket --> positive
He does not like cricket --> negative
They are not playing football --> negative
We have not completed the work --> negative
```